10. A technician enters an invalid item price

The technician attempts to save a price outside the allowed range during maintenance mode.

Objective	• Prevent an invalid price from replacing an item's existing price.	
Results	• The controller rejects the invalid entry, preserves the existing price, and allows the technician to correct the entry or cancel.	
Actors	• Technician, who understands the maintenance controls; vending machine controller.	
Related use cases	• Use Case 7: Initial setup. Use Case 8: Changing an item's price.	
Trigger	• The technician attempts to save a price below $0.05 or above $9.99.	
Stakeholders	• The machine owner and customers, who depend on accurate item prices and charges.	
Resources	• Programmed SoC Series V board, and Quartus.	
Assumptions	• A working setup with the seven-segment display, working money switches, working RAM set up for Items and the working keypad as well.	
Post-condition	• For example, Location 11 still costs $1.35 after the invalid entry is rejected. Other stored prices remain unchanged, and the technician can correct the entry or cancel.	
Frequency	• Whenever a technician enters a price outside the permitted range.	
Mission Priority	• High. Invalid prices must not be used for customer purchases.	
Safety Priority	• Should hold no safety concern	
Flow of events	**Step #**	**Action**
	1.	The technician confirms that no customer transaction is active and enters maintenance mode.
	2.	The controller indicates that maintenance mode is active.
	3.	The technician selects location 11, and the controller displays its current price of $1.35.
	4.	The technician enters a new price of $0.00.

	5.	The technician uses the designated save control.
	6.	The controller checks the entered price and determines that $0.00 is below the minimum price of $0.05.
	7.	The controller rejects the entry, provides a visible invalid-price indication, and keeps location 11's stored price at $1.35.
Alternate Events	**Step #**	**Action**
	4A	If the technician enters $0.04, the controller rejects it because it is below the minimum price.
	4B	If the technician attempts to enter $10.00, the controller rejects the out-of-range entry without saving it or silently dropping a digit.
	4C	If the technician enters exactly $0.05 or $9.99, the controller accepts the price when the technician saves because both limits are included in the allowed range.
	7A	The technician replaces the invalid entry with $1.50 and saves. The controller validates the new value and updates only location 11.
	7B	The technician cancels the unsaved change and exits maintenance mode. Location 11 remains priced at $1.35.
Hazard/ Result /Mitigation	• If case is used incorrectly, the entire system is likely incorrect or not working to specifications.	
Requirements from Use Case	• Reject prices below $0.05 or above $9.99. • Preserve the existing item price and all other prices when an entry is rejected. • Provide a visible indication that the entered price is invalid. • Allow the technician to correct an invalid entry and try again, or cancel without changing the stored price. • Accept valid prices at both boundaries, including $0.05 and $9.99, when saved. • Prevent excess digits from being silently discarded and saved as an unintended price.	